World Health Spending & Sanitary Performance

Does investing more in health guarantee better outcomes? An analysis of **200+ countries over 20 years** (1999–2019)

PYTHON | SQL MYSQL | POWER BI

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Context & Objective

Business Question

To what extent does health spending determine performance on avoidable mortality, life expectancy, and infectious disease control — and which countries show the most critical gaps between investment and outcomes?

Context

Global health systems face growing pressure to justify public resource allocation. Some countries achieve excellent results with modest budgets, while others, despite high investment, display alarming indicators.

Decision Purpose

Provide decision-makers with a prioritization tool to direct health investments where they will produce the greatest human impact.

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Dataset & Analytical Pipeline

WHO Global Health Observatory

6,650
Observations

200+
Countries

21
Years (1999–2019)

11
Final Variables

Key Variables: health_spending_pct_gdp · life_expectancy_years · avoidable_mortality_index · infectious_disease_index · health_roi_score

Analytical Pipeline

Python

Quality audit, cleaning, feature engineering

SQL MySQL

10 decisional analytical queries

Power BI

3-page dashboard, 9 visuals, 20+ DAX measures

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Python Step: Cleaning & Feature Engineering

Quality Audit

Missing Values

Detected on 9 columns → Median imputation by country

Outliers

Detected via IQR method → Capping without deletion (real-world conditions preserved)

6 Business Variables Created

Variable	Description
health_spending_category	Low / Medium / High per WHO thresholds
avoidable_mortality_index	Score 0→1: infant + maternal + under-5 mortality
infectious_disease_index	Score 0→1: HIV + tuberculosis + malnutrition
health_roi_score	Life expectancy / health spending
life_expectancy_category	Critical / Low / Medium / High
performance_gap_flag	Critical Gap vs Normal

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Key Finding #1: Under-Investment as the Primary Driver

🔴 **Low (<5% GDP)**

71 countries

Avg: 3.1%

🟡 **Medium (5–9% GDP)**

97 countries

Avg: 6.8%

🟢 **High (≥10% GDP)**

38 countries

Avg: 12.4%

Critical finding: The 10 countries with the lowest life expectancy are ALL in the Low spending category and located in Sub-Saharan Africa.

01

Central African Republic

53.1 years — Low spending

02

Chad

54.2 years — Low spending

03

Nigeria

54.8 years — Low spending

04

Sierra Leone

55.0 years — Low spending

05

Lesotho

55.3 years — Low spending

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Key Finding #2: Efficient Models at Low Cost

Some countries achieve excellent outcomes with modest budgets — proof that **well-directed spending toward prevention and primary care** generates life expectancy comparable to countries investing 2–3× more.

Country	Spending	Life Expectancy	ROI Score
Bangladesh	2.8% GDP	71.2 years	25.4
Sri Lanka	3.5% GDP	74.8 years	21.4
Egypt	4.9% GDP	71.8 years	14.7
Morocco	5.2% GDP	72.4 years	13.9
Vietnam	5.7% GDP	73.6 years	12.9

Key Insight

Bangladesh achieves 71.2 years of life expectancy spending only **2.8% of GDP** — a ROI score of 25.4, among the highest globally.

These models are **replicable** and represent the most actionable lever for low-resource countries.

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Key Finding #3: The Critical Investment/Results Gap

Some countries spend heavily but achieve poor outcomes — a structural misalignment between investment volume and health system efficiency.

Lesotho

11.2% GDP spent · 55.3 years life expectancy · ⚠️ **Critical Gap**

Botswana

10.3% GDP spent · 64.2 years life expectancy · ⚠️ **Critical Gap**

South Africa

8.7% GDP spent · 63.1 years life expectancy · ⚠️ **Critical Gap**

Namibia

8.1% GDP spent · 63.8 years life expectancy · ⚠️ **Critical Gap**

Probable cause: High uncontrolled HIV prevalence + inequalities in access to care + poorly oriented spending structure (curative vs. preventive)

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Key Finding #4: Countries with a Double Health Burden

These countries face **simultaneously high avoidable mortality AND intense infectious disease pressure** — making them the absolute priority targets for international humanitarian interventions.

Central African Republic

Mortality Index: **0.891**

Infectious Index: **0.823**

Spending: Low

Chad

Mortality Index: **0.876**

Infectious Index: **0.798**

Spending: Low

Sierra Leone

Mortality Index: **0.854**

Infectious Index: **0.771**

Spending: Low

Nigeria

Mortality Index: **0.831**

Infectious Index: **0.812**

Spending: Low

South Sudan

Mortality Index: **0.819**

Infectious Index: **0.756**

Spending: Low

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Key Findings #5 & Temporal Evolution

Finding #5: Spending Controls Infectious Diseases

Health spending is a **direct lever on infectious disease pressure**. High-spending countries control infectious diseases **4× better** than low-spending countries.

18%

High Spending

38 countries — avg infectious index 0.18

34%

Medium Spending

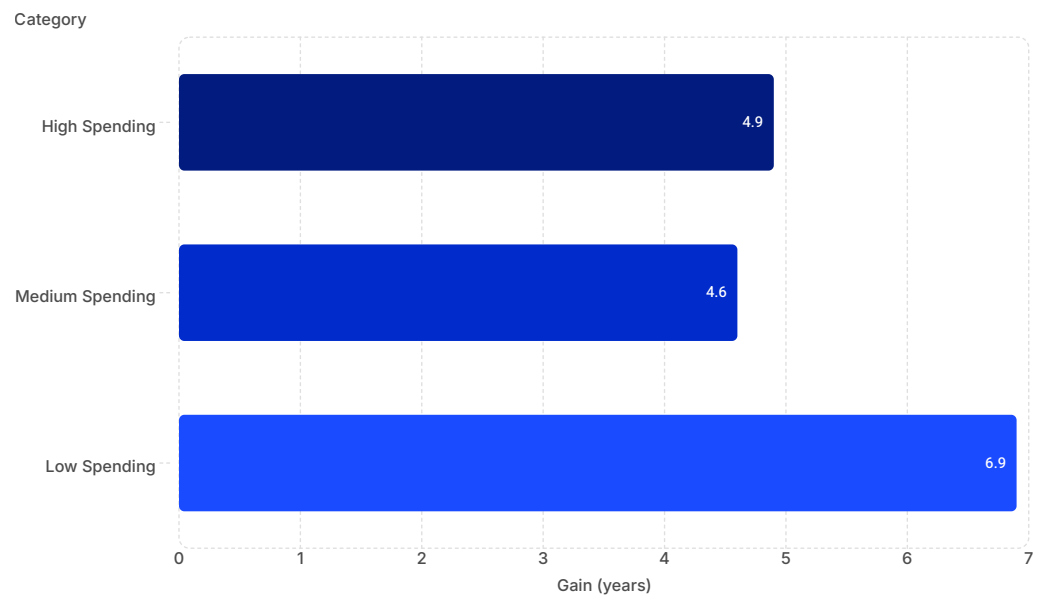
97 countries — avg infectious index 0.34

67%

Low Spending

71 countries — avg infectious index 0.67

Life Expectancy Gain 1999–2019



Catch-up paradox: Low-spending countries recorded the largest absolute gain (+6.9 years) — but their starting level remains dramatically low (54.3 years in 1999).

ROI Champions (1999–2019)

1 **Ethiopia**
+7.6 ROI points

2 **Rwanda**
+7.2 ROI points

3 **Bangladesh**
+6.8 ROI points

4 **Cambodia**
+6.5 ROI points

5 **Nepal**
+6.3 ROI points

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Power BI Dashboard Structure

3	9	20+	5
Dashboard Pages	Visuals	DAX Measures	Slicers

Page 1 — Global View

- KPI Cards: countries analyzed, avg spending, life expectancy, critical gap countries
- Donut chart: spending distribution
- Top 15 ROI countries
- World choropleth map

Page 2 — Mortality & Infectious Diseases

- KPI Cards: synthetic indices
- Bar chart: Top 20 avoidable mortality
- Scatter plot: double burden
- Conditional table: priority countries

Page 3 — Health ROI & Evolution

- KPI Cards: ROI metrics
- Line chart: temporal evolution
- Scatter plot: spending vs life expectancy
- Bar chart: ROI Low vs High comparison

Slicers: Year · Country · Spending Level · Life Expectancy Performance · Critical Gap

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